Impact of Delivery Delays on Customer Satisfaction

Data Storytelling & Statistical Validation

Project: E-Commerce Order Fulfilment Efficiency Analysis

Business Problem

Business Context

- Delivery performance strongly influences customer satisfaction in e-commerce.

Key Question

- Do late deliveries significantly reduce customer review scores?

Objective

- Validate impact using statistical hypothesis testing.

Business Impact

- Lower ratings
- Customer churn risk
- Negative brand perception
- Operational improvement opportunity

Dataset Overview

Total Orders
96,353

Late Orders
6,409

On-Time Orders
89,944

Key Dataset Facts

- Late delivery rate: **6.6%**
- Analysis based on **96K delivered orders**
- Includes customer review ratings

Hypothesis Testing

- H0: No significant difference in ratings
- H1: Late deliveries have lower ratings
- Test: Independent T-Test
- Significance level: **0.05**

Key Results

Late Delivery Rating
2.27

On-Time Delivery Rating
4.29

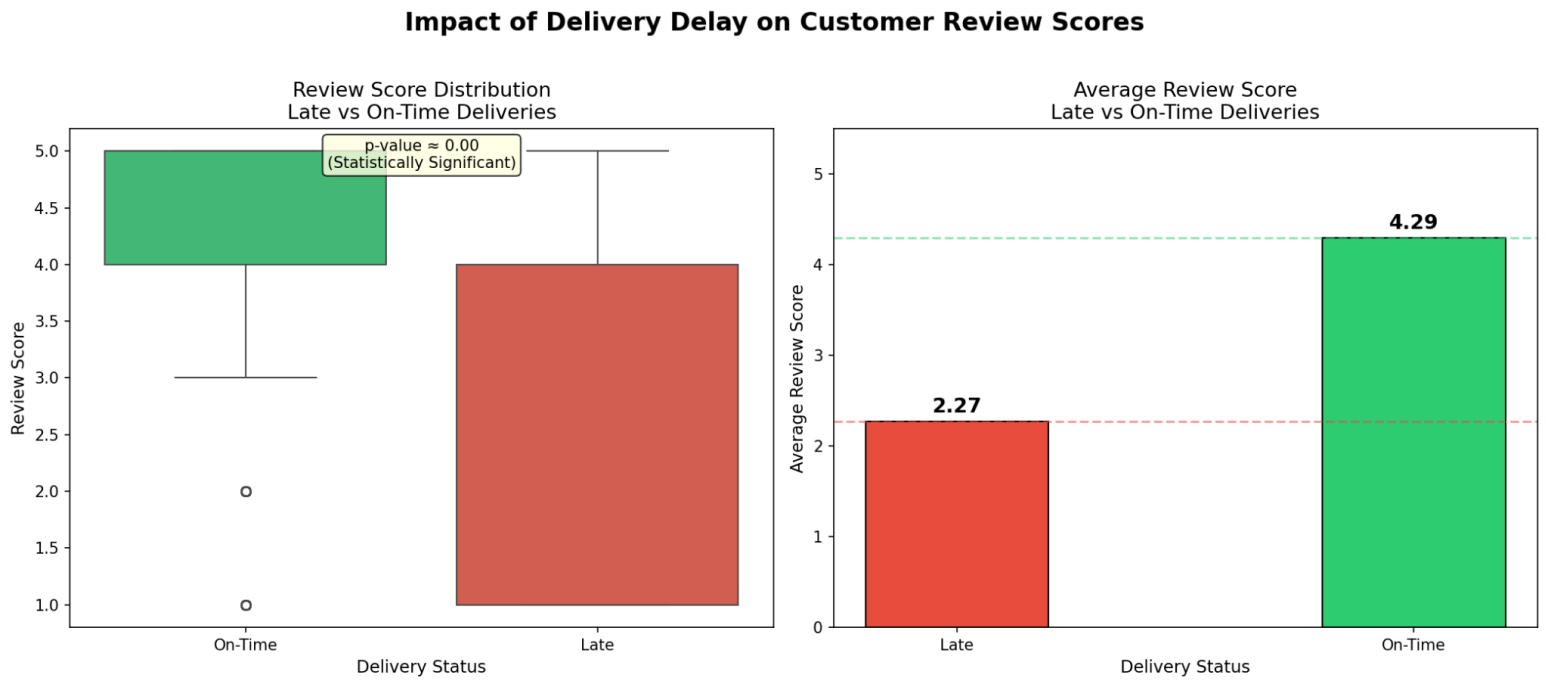
Rating Gap
2.02

Key Insight:
On-time deliveries receive ~2 point higher ratings.
Statistical Result:
Difference is statistically significant ($p < 0.05$).

Statistical Validation

- T-Statistic: **-132.07**
  P-Value: **< 0.0001**
- Decision: **Reject Null Hypothesis**
- Conclusion:
  Late deliveries significantly reduce ratings.

Key Insight Visualization



Business Insights

- Late deliveries damage customer experience
- Delivery reliability drives satisfaction
- Logistics efficiency impacts retention

Recommendations

- Improve logistics
- Predict delays
- Notify customers
- Offer compensation
- Monitor partners

Conclusion

- Late deliveries reduce ratings by **2.02 points**
- Result statistically significant (**$p < 0.05$**)
- Delivery performance directly affects satisfaction
- Improving logistics can improve retention.